

THE UNITED REPUBLIC OF TANZANIA
PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
DARES SALAAM REGION
FORM FOUR - MOCK EXAMINATION - 2026

CHEMISTRY 1

CODE:032/1

TIME:3:00Hours

Thursday 14th May, 2026 A.M

INSTRUCTIONS

1. This paper consists of sections **A, B** and **C** with a total of **eleven (11)** questions.
2. Answer **all** questions in sections A and B and two (2) questions from section C.
3. Section A carries **sixteen (16)** marks, section B carries **fifty four (54)** marks and section C carries **thirty (30)** marks.
4. All writings must be in **blue** or **black** ink with exception of diagrams which must be in pencil.
5. Non-programmable calculators and mathematical tables may be used.
6. Write your **Examination Number** on every page of your answer booklet(s).
7. The following constants may be used:
Avogadro's number = 6.02×10^{23}
Atomic masses
H = 1, C = 12, O = 16, Na = 23, Al = 27, S = 32, Mg = 24, Cu = 64,
Ag = 108, Zn = 65

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This paper consists of 5 printed pages

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SECTION A (16 marks)

Answer **all** questions in this section.

1. For each of the items (i) – (v), choose the most correct answer from the given alternatives and write its letter in the answer booklet provided.
- (i) Which of the following groups consists of home care products?
- A. Yeast, plastics and disinfectants
 - B. Paints, air fresheners and petrol
 - C. Rubbers, clothes and soap
 - D. Detergents, disinfectants and air fresheners
 - E. Clothes, soap and stone
- (ii) A farmer was looking for a chemical compound that he can use as liming. Which among the following be his best choice?
- A. Calcium hydroxide and calcium oxide
 - B. Magnesium oxide and sodium hydroxide
 - C. Calcium oxide and sodium hydroxide
 - D. Magnesium hydroxide and sodium oxide
 - E. Sodium oxide and potassium hydroxide
- (iii) 2 moles of sodium hydroxide are dissolved in water, what is the total number of moles of ions produced?
- A. 1
 - B. 2
 - C. 3
 - D. 4
 - E. 5
- (iv) Which compound among the following causes temporary hardness of water when dissolved in water?
- A. NaCl
 - B. CaCl_2
 - C. MgSO_4
 - D. $\text{Mg}(\text{HCO}_3)_2$
 - E. $\text{Pb}(\text{NO}_3)_2$
- (v) Why should access to safety equipment never be blocked?
- A. It is a simple law
 - B. The equipment is used daily
 - C. Is essential in case of an accident
 - D. People need space to move in the laboratory
 - E. It is a law of energy
- (vi) There is a moment when even the atom or ions of different elements may have the same number of subatomic particles as discovered by J.J Thomson. Which among the following neutral atoms have the same number of particles as Mg^{2+}
- A. Magnesium
 - B. Neon
 - C. Sodium
 - D. Argon
 - E. Potassium

- (vii) Why is calcium oxide useful during preparation of ammonia?
- It removes water from the reaction
 - It prevents the removal of water in the reaction
 - It removes nitrogen gas from the reaction
 - It prevents the loss of nitrogen from the reaction
 - It increases the production of ammonia
- (viii) Gaseous fuel include
- Water gas and petrol
 - Water gas and kerosene
 - Water gas and producer gas
 - Coke and producer gas
 - Coke and natural gas
- (ix) Lowering activation energy of a reaction requires
- High pressure
 - Adding catalyst for the reaction
 - High temperature of reaction
 - High pressure and concentration of reaction
 - Both pressure and temperature rise
- (x) Which of the following is NOT an organic substance
- Cooking oil
 - Rubber
 - Car tires
 - Baking powder
 - Plastics

2. Match the metal nitrate in **list A** with the corresponding products formed on heating in **list B** by writing the letter of the correct response besides the item number in the answer booklet provided

LIST A	LIST B
(i) Copper	A. Sodium nitrate and oxygen
(ii) Mercury	B. Dinitrogen oxide and water
(iii) Sodium	C. Mercury, nitrogen dioxide and oxygen
(iv) Aluminium	D. Sodium nitrite and oxygen
(v) Ammonium	E. Aluminium oxide, nitrogen dioxide and oxygen
(vi) Silver	F. Mercury, dinitrogen oxide and oxygen
	G. Silver, nitrogen dioxide and oxygen
	H. Copper oxide, nitrogen dioxide and oxygen

SECTION B (54 marks)

Answer **all** questions in this section

3. (a) Explain the following
- All alkalis are bases but not all bases are alkalis
 - Acidity of a base and basicity of an acid are not the same
 - Neutralisation can treat both soil and waste

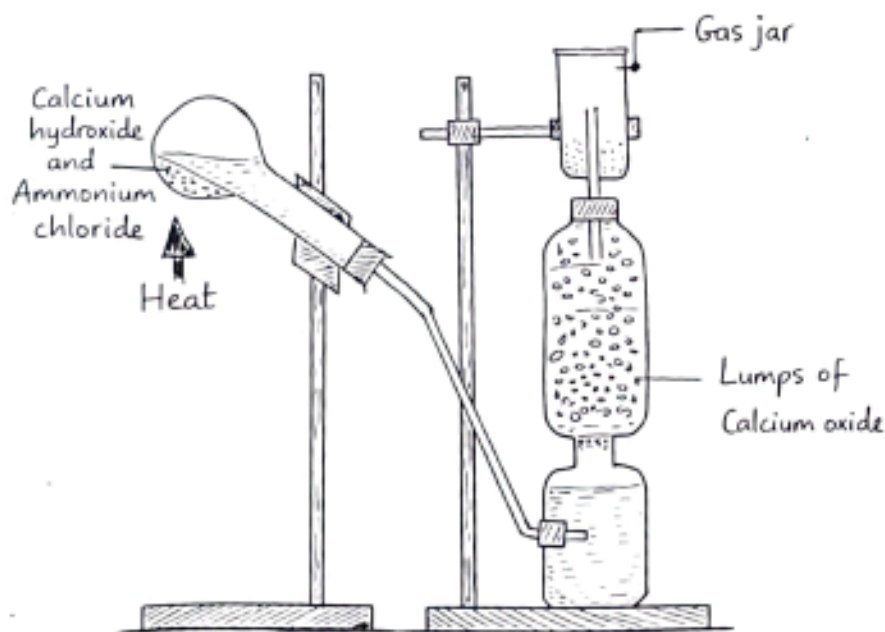
- (b) When an acid found in rotten fruits such as grapes and pineapples come into contact with ethanol (an alcohol) in presence of concentrated sulphuric acid a sweet-smelling compound is formed. Name the reactants and product formed
4. (a) The following are names of laboratory apparatus: Tripod stand, pipette, conical flask, wire gauze, Bunsen burner, combustion tube, burette, white tile. Identify four apparatus used in volumetric analysis.
- (b) Mr. Othman and Alvis were arguing on the usefulness of volumetric analysis in our daily life. As form four student explain two challenges and educate them on the usefulness of volumetric analysis. Give five (5) points.
5. (a) (i) Explain the concept of water of crystallisation as used in salts.
 (ii) Which of the following compounds has greater percentage of water of crystallisation; $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ and $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$
- (b) How many
 (i) Moles are there in 20g of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
 (ii) Electrons are needed to discharge 5g of copper by electrolysis
 (iii) Aluminium ions will be present in solution if 3g of aluminium sulphate are completely dissolved in water
6. (a) Study the following ions with their neutral atoms: K^+ , Cl^- , K and Cl Why is K^+ slightly smaller than K atom and Cl^- is slightly larger than Cl atom
- (b) (i) Nitrogen and hydrogen atoms combine by sharing equal number of electrons to form ammonia gas molecule. Illustrate the formation of a compound formed by using electron diagram.
 (ii) Give three properties of a compound formed above
7. (a) The reaction of acid and bases is applied in various useful situations. With the aid of balanced chemical equation where necessary, describe any four (4) usefulness of the reaction.
- (b) 2.91 g of a monobasic acid, HX were dissolved in water and made up to 250cm^3 solution. The solution was titrated with 0.108M sodium hydroxide solution. 25cm^3 of the sodium hydroxide required 22.5 cm^3 of the HX solution for complete neutralisation. The equation for the reaction is:
 $\text{HX} + \text{NaOH} \longrightarrow \text{NaX} + \text{H}_2\text{O}$
 (aq) (aq) (aq) (l)
 (i) Calculate the concentration in grams per litre and moles per litre.
 (ii) Find the molar mass of HX.
8. (a) (i) Briefly explain the term metallurgy.
 (ii) Why when a piece of magnesium ribbon is added into a solution of dilute hydrochloric acid, some gas bubbles are observed but when a piece of copper wire is used instead of magnesium no bubbles are observed.

- (b) (i) Extraction of metals by electrolytic reduction is a type of electrolysis, give reason.
- (ii) What happens in the Downs cell if the molten sodium is mixed with chlorine at very high temperature during electrolytic extraction of sodium.

SECTION C (30 marks)

Answer two (2) questions in this section

9. (a) What the fire and fire fighting concept mean in chemistry
- (b) You have been assigned to explain the classes of fire. Use the following burning materials to explain the respective classes of fire and give one appropriate extinguisher for each class. "oil cooking plate, magnesium powder, and kerosene
- (c) A young scientist was shocked to see a dangerous fire on a large building. He wanted to stop it using dry powder extinguisher that was around the building. Unfortunately he did not know the procedures on how to use it. As a chemist, guide him systematically on how to use it.
10. The following figure show a set-up that can be used to prepare a gas X in the laboratory



- (a) Identify gas X
- (b) Write an equation to show the formation of the gas
- (c) Why is the gas not collected over water? Explain
- (d) Describe a chemical test that can be used to confirm gas X
- (e) With the aid of an equation, explain what would happen if a dry sample of gas X is passed over heated copper(II) oxide.
11. Pollution from bars, concert, radios and TV is among the problems in cities around the world. It poses a number of health problems like hearing loss, sleep disturbances, children diseases and dysfunctions, hypertension and cardiovascular dysfunction. Discuss the causes and measures to control that kind of pollution.

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